UART CLI Homework — Test Report

Student: Alexandra Holoborodko Date: 5.10.26

Objective: find robustness defects without inspecting the source code first.

1. Test Execution

TC ID	Req	Input / Sequence	Technique	Expected	Actual	Evidence	Result	Bug ID
TC-01	R01	Any valid command from supported by CLI (help, status, set speed <value>, set timeout <value>, led on, led off, save, reset)	Boundary Value Analysis		1.help->commands list 2.status->system actual status: SPEED=120 TIMEOUT=10 LED=OFF CONFIG=SAVED 3.led on->OK, led is on 4.led off->OK, led is off 5.save-> OK: configuration saved 6.reset->OK: reset 7.set speed 70->OK: speed=70 8.set timeout 50->OK: timeout=50		PASS	
TC-02	R01 R07	Any invalid command: test command; stat; set speed<value> (without space), тесткоманда [Cyrillic "test command"]	Boundary Value Analysis / Negative Testing	Unknown or incomplete commands must return a controlled ERROR	invalid command: test command; stat 1. Status: SPEED=60 TIMEOUT=50 LED=OFF CONFIG=DIRTY -> ERROR: unknown command 2. Status: SPEED=4 TIMEOUT=50 LED=OFF CONFIG=SAVED ->ERROR: unknown command invalid command: set speed5 (without space) CONFIG=DIRTY OK: speed=5 CONFIG=SAVED > OK: speed=5 invalid command: тесткоманда [Cyrillic "test command"] CONFIG=DIRTY -> OK CONFIG=SAVED -> OK		FAIL	Bug_02
TC-03	R01 R09	1.Enter any invalid command: Test command; stat; set timeout<value>(without space) 2.Enter Status command	Boundary Value Analysis/ Negative Testing	Invalid commands must not change device configuration	Invalid commands are not changed device configuration		PASS	
TC-04	R02	set speed <value>: 0,1, 254, 255	Boundary Value Analysis	`speed` accepts integer values from 0 to 255 inclusive	`speed` accepts integer values from 0 to 255 inclusive		PASS	
TC-05	R03	set speed <value>: -1, 256	Boundary Value Analysis	Values outside 0..255 must return an ERROR and must not modify speed	set speed <-1>: ERROR: speed out of range set speed <256>: OK: speed=255		FAIL	Bug_03
TC-06	R02	set speed <value>: 0.25; A, @#4, empty value	Negative Testing	Controlled ERROR should be returned	CONFIG=DIRTY ERROR: invalid speed CONFIG=SAVED ERROR: invalid speed		PASS	
TC-07	R04	set timeout <value>: 1,2,59,60	Boundary Value Analysis	`timeout` accepts integer values from **1 to 60 inclusive**	`timeout` accepts integer values from **1 to 60 inclusive**		PASS	
TC-08	R05	set timeout <value>:0,61	Boundary Value Analysis	Values outside 1..60 must return an ERROR and must not modify timeout	CONFIG=SAVED CONFIG=DIRTY OK: timeout=0 timeout=61 -> ERROR: timeout out of range		FAIL	Bug_04
TC-09	R04	set timeout <value>: 0.25; A, @#4, empty value	Negative Testing	Controlled ERROR should be returned	CONFIG=SAVED CONFIG=DIRTY ERROR: invalid timeout		PASS	
TC-10	R06	Uppercase: HELP, STATUS, SET SPEED 4, SET TIMEOUT 8, LED ON, LED	Negative Testing	Commands are case-sensitive	CONFIG=SAVED CONFIG=DIRTY ERROR: unknown command		PASS	Bug_06

		OFF, RESET, SAVE Camelcase: LeD oN			STATUS->accepted			
TC-11	R07 R08	Special-character-only input: #@*&	Negative Testing	Special-character-only input must not be treated as a valid command; Unknown or incomplete commands must return a controlled ERROR	CONFIG=SAVED CONFIG=DIRTY > OK		FAIL	Bug_05
TC-12	R10	CLI input length: 0, 1,39, 40 characters	Input Length Testing	Maximum CLI input length is 40 characters	ERROR: empty command ERROR: unknown command		PASS	
TC-13	R11	CLI input length is >40 characters: 41	Input Length Testing /Negative Testing	Any line longer than 40 characters must return `ERROR: command too long`	*** SYSTEM FAULT *** Reason: input line boundary failure rst:0xc (SW_CPU_RESET) Rebooting...		FAIL	Bug_01
TC-14	R12	CLI input length is >40 characters: 256	Stress Input	Long input must not cause reboot, hang or parser corruption	*** SYSTEM FAULT *** Reason: input line boundary failure rst:0xc (SW_CPU_RESET) Rebooting...		FAIL	Bug_01
TC-15	R13	1.Enter valid command: help 2.Enter invalid command: test	State / Sequence Testing	Invalid input must behave the same regardless of previous valid commands	> ERROR: unknown command		PASS	
TC-16	R14	1. Set speed/timeout = state is dirty 2.Enter invalid command	State / Sequence Testing	A DIRTY configuration state must not make invalid commands dangerous	A DIRTY configuration state is not made invalid commands dangerous		PASS	
TC-17	R15	1.Enter set timeout <61> 2.Enter status	State / Sequence Testing	After every validation error, `status` must remain available	After every validation error, `status` is available		PASS	
TC-18	R16	1.Enter valid/invalid command; input	State / Sequence Testing	User input must never cause an unexpected reboot or crash	See TC-13 & TC-14		FAIL	Bug_01
TC-19 (added after webinar)	R16	1. Enter set speed 200 2.Enter set	State / Sequence Testing	User input must never cause an unexpected reboot or crash	*** SYSTEM FAULT *** Reason: parser state mismatch while configuration is dirty rst:0xc (SW_CPU_RESET) Rebooting...		FAIL	Bug_07

2. Boundary / Length Analysis

Speed boundary values tested: -1, 0,1, 254, 255, 256

Timeout boundary values tested: 0, 1,2,59,60, 61

Input lengths tested: 0, 1,39, 40, 41, 256

Exact behavior-change boundary: 41 - SYSTEM FAULT

Observed result at N-1 / N / N+1:

- N-1=40 system returned ERROR: unknown command
- N=41 SYSTEM FAULT
- N+1=42 SYSTEM FAULT

3. Defects Found

BUG-01

Title: SYSTEM FAULT after entering CLI input >40 symbols

Steps:

1. Turn on Wokwi simulation

2. Open Serial Monitor

3. Enter CLI input >40 symbols (helphelphelphelphelphelphelphelphelphelphelp)

4. Check output

Expected:

`ERROR: command too long

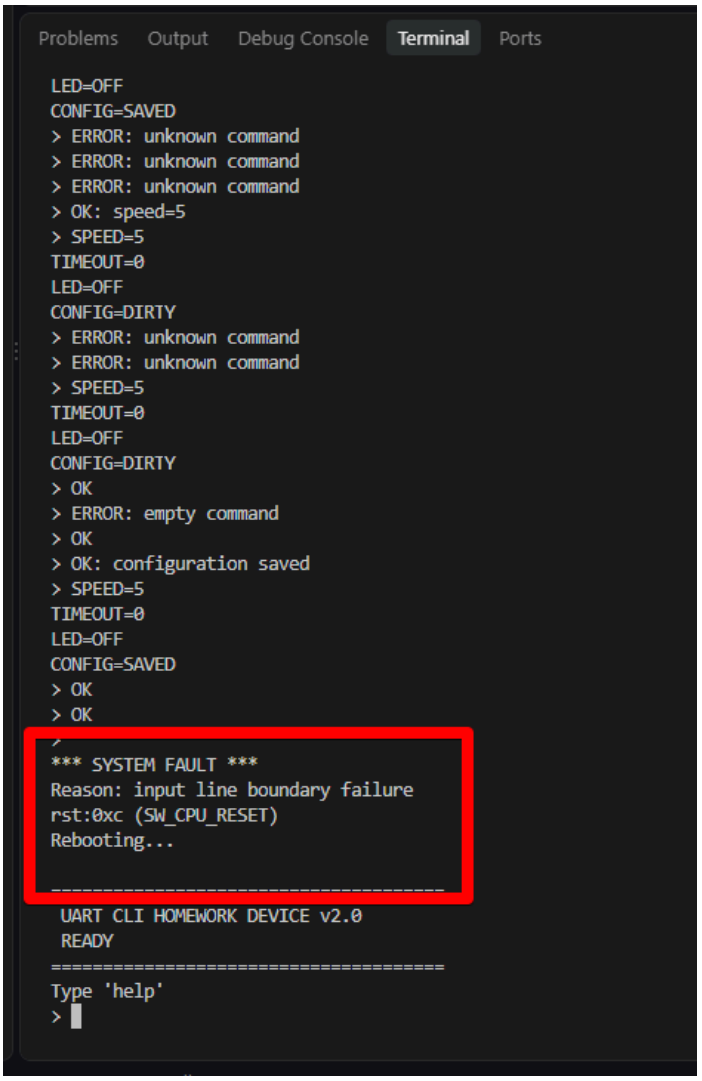
Long input must not cause reboot, hang or parser corruption

Actual: SYSTEM FAULT, system is rebooting

Severity: Critical

Reproducibility: always

Evidence:



BUG-02

Title: Error is not displayed after CLI input of invalid command

Steps:

1. Turn on Wokwi simulation

2. Open Serial Monitor

3. Enter CLI input set speed5 (without space); тесткоманда [Cyrillic for "test command"]

4. Check output

Expected: ERROR: unknown command

Actual:

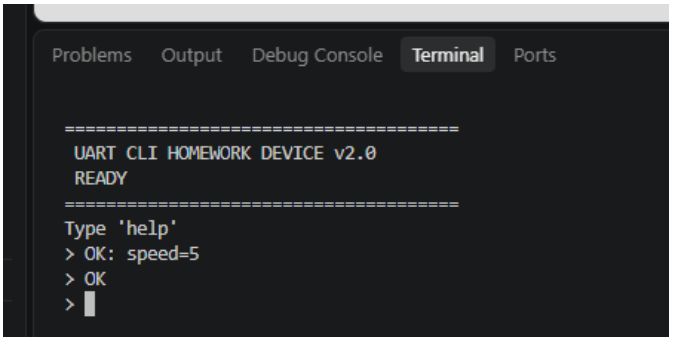
set speed5 ->OK: speed=5

тесткоманда [Cyrillic for "test command"] ->OK

Severity: Major

Reproducibility: always

Evidence:



BUG-03

Title: System saves speed value 255 after entering command <set speed 256>

Steps:

1. Turn on Wokwi simulation
2. Open Serial Monitor
3. Enter CLI input <set speed 256>
4. Check output

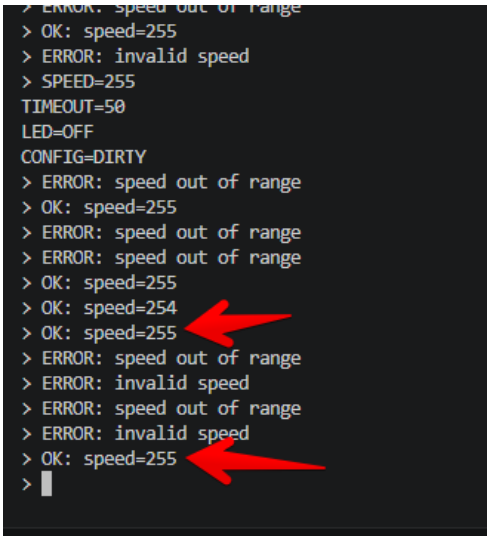
Expected: ERROR is returned and speed is not modified

Actual: OK: speed=255; config=dirty

Severity: major

Reproducibility: always

Evidence:



BUG-04

Title: Error is not returned and timeout is modified after entering command <set timeout 0>

Steps:

1. Turn on Wokwi simulation

2. Open Serial Monitor

3. Enter CLI input <set timeout 0>

4. Check output

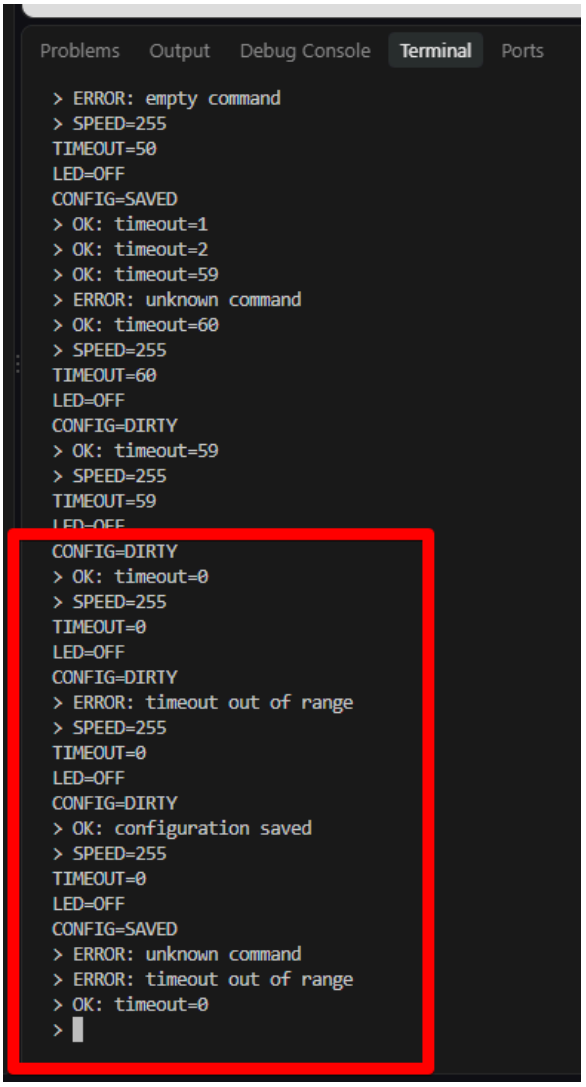
Expected: ERROR is returned and timeout is not modified

Actual: OK: timeout=0

Severity: major

Reproducibility: always

Evidence:



BUG-05

Title: OK is returned after entering special character only

Steps:

1. Turn on Wokwi simulation

2. Open Serial Monitor

3. Enter CLI input special character only: #@*&

4. Check output

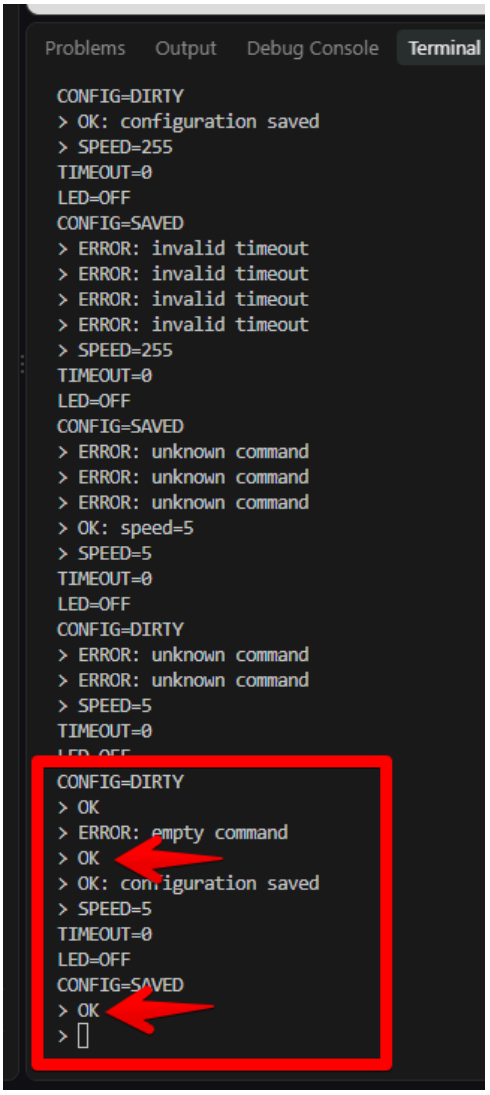
Expected: ERROR is returned

Actual: >OK

Severity: major

Reproducibility: always

Evidence:



BUG-06

Title: Uppercase command STATUS is acceptable

Steps:

1. Turn on Wokwi simulation

2. Open Serial Monitor

3. Enter CLI input STATUS (upper case)

4. Check output

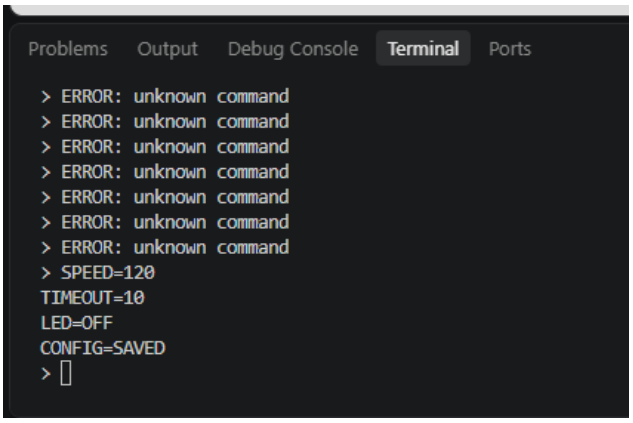
Expected: ERROR: unknown command, command is case sensitive

Actual: STATUS is accepted

Severity: major

Reproducibility: always

Evidence:



BUG-07

Title: SYSTEM FAULT after entering incomplete command <set> if config is DIRTY

Steps:

1. Turn on Wokwi simulation
2. Open Serial Monitor
3. Enter CLI input <set speed 200>/<set timeout 45> (config is dirty)
4. Enter CLI input <set>
5. Check output

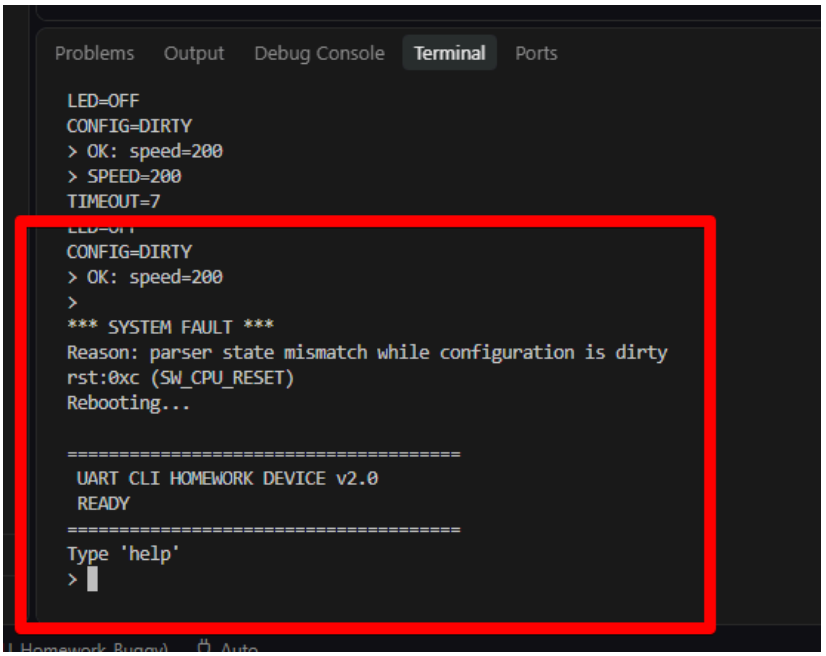
Expected: ERROR is returned

Actual: SYSTEM FAULT, system is rebooting

Severity: critical

Reproducibility: always

Evidence:



4. QA Conclusion

Happy paths are passed, but robustness requirement is not met. Found two critical behaviour: system is rebooting on 41 symbol edge (Input Length Testing) and after entering incomplete command <set> if config is DIRTY (State / Sequence Testing). Other five issues are major and also needed to be fixed: incorrect processing of speed=256

and timeout=0 (Boundary Value Analysis); case sensitive requirement is broken for STATUS command and special symbols input only (parsing); OK is returned for some invalid commands.
Build is not approved for release until all critical and major issues are fixed.